



PROJECT CONTROLS & COST CONTROL

FORMULA SHEET



PLANNING • SCHEDULING • COST ENGINEERING • EARNED VALUE • CONSTRUCTION MANAGEMENT

1. PROJECT PROGRESS

1. Percent Complete

$$\% \text{ Complete} = \frac{\text{Actual Quantity}}{\text{Total Quantity}} \times 100$$

2. Physical Progress

$$\text{Physical Progress} = \frac{\sum (\text{Weight}_i \times \text{Progress}_i)}{\sum \text{Weight}_i}$$

3. Remaining Progress

$$\text{Remaining} = 100 - \% \text{ Complete}$$



2. EARNED VALUE MANAGEMENT (EVM)



1. Planned Value (PV)

$$PV = \text{Budget} \times \text{Planned \% Complete}$$

2. Earned Value (EV)

$$EV = \text{Budget} \times \text{Actual \% Complete}$$

3. Actual Cost (AC)

$$AC = \sum \text{Actual Expenditure}$$

4. Budget at Completion (BAC)

$$BAC = \sum \text{Budget}$$



3. VARIANCE ANALYSIS



1. Cost Variance (CV)

$$CV = EV - AC$$

2. Schedule Variance (SV)

$$SV = EV - PV$$

3. Cost Variance %

$$CV\% = \frac{CV}{EV} \times 100$$

4. Schedule Variance %

$$SV\% = \frac{SV}{PV} \times 100$$



4. PERFORMANCE INDICES



1. Cost Performance Index (CPI)

$$CPI = \frac{EV}{AC}$$

2. Schedule Performance Index (SPI)

$$SPI = \frac{EV}{PV}$$

3. Critical Ratio (CR)

$$CR = CPI \times SPI$$



5. FORECASTING



1. Estimate at Completion (EAC)

$$EAC = \frac{BAC}{CPI}$$

$$(\text{or}) EAC = AC + (BAC - EV)$$

2. Estimate to Complete (ETC)

$$ETC = EAC - AC$$

3. Variance at Completion (VAC)

$$VAC = BAC - EAC$$

4. To Complete Performance Index (TCPI)

$$TCPI = \frac{BAC - EV}{BAC - AC}$$



6. COST CONTROL



1. Cost Growth

$$\text{Cost Growth} = \frac{\text{Final} - \text{Budget}}{\text{Budget}} \times 100$$

2. Unit Cost

$$\text{Unit Cost} = \frac{\text{Total Cost}}{\text{Quantity}}$$

3. Cost Efficiency

$$\text{Efficiency} = \frac{\text{Budgeted Cost}}{\text{Actual Cost}}$$

4. Budget Utilization

$$\text{Budget Used} = \frac{\text{Actual}}{\text{Budget}} \times 100$$



7. CASH FLOW



1. Net Cash Flow

$$\text{Cash}_{\text{Net}} = \text{Income} - \text{Expenses}$$

2. Cumulative Cash Flow

$$\text{Cumulative} = \sum \text{Monthly Cash Flow}$$

3. Burn Rate

$$\text{Burn Rate} = \frac{\text{Actual Cost}}{\text{Time}}$$

4. Monthly Spending Rate

$$\text{Rate} = \frac{\text{Monthly Cost}}{\text{Working Days}}$$



8. PROCUREMENT



1. Commitment

$$\text{Commitment} = \text{PO} + \text{Subcontracts}$$

2. Outstanding Commitment

$$\text{Outstanding} = \text{Commitment} - \text{Paid}$$

3. Retention Balance

$$\text{Retention} = \text{Certified Value} \times \text{Retention \%}$$

4. Advance Recovery

$$\text{Recovery} = \text{Invoice} \times \text{Recovery \%}$$



9. PRODUCTIVITY



1. Labour Productivity (LP)

$$LP = \frac{\text{Output}}{\text{Labour Hours}}$$

2. Equipment Productivity (EP)

$$EP = \frac{\text{Output}}{\text{Machine Hours}}$$

3. Resource Utilization (RU)

$$RU = \frac{\text{Actual Hours}}{\text{Available Hours}} \times 100$$

4. Crew Efficiency

$$\text{Efficiency} = \frac{\text{Earned Hours}}{\text{Actual Hours}} \times 100$$



10. PLANNING & SCHEDULING



1. Total Float (TF)

$$TF = LS - ES = LF - EF$$

2. Free Float (FF)

$$FF = ES_{\text{next}} - EF$$

3. Activity Duration

$$\text{Duration} = \text{Finish} - \text{Start}$$

4. Remaining Duration

$$\text{Remaining} = \text{Original} - \text{Elapsed}$$



11. RISK MANAGEMENT



1. Expected Monetary Value (EMV)

$$EMV = \text{Probability} \times \text{Impact}$$

2. Contingency

$$\text{Contingency} = \sum EMV$$

3. Risk Exposure

$$\text{Exposure} = \text{Probability} \times \text{Consequence}$$



12. CONTRACTS & FINANCIAL KPIs

A. CONTRACTS

1. Liquidated Damages (LD)

$$LD = \text{Rate} \times \text{Delay Days}$$

2. Retention

$$\text{Retention} = \text{Invoice} \times \text{Retention \%}$$

3. Price Escalation (PE)

$$PE = \text{Contract Value} \times \text{Escalation Index}$$

4. Change Order %

$$CO\% = \frac{\text{Variation}}{\text{Original Contract}} \times 100$$

5. Contract Completion %

$$\text{Completion \%} = \frac{\text{Certified Value}}{\text{Contract Value}} \times 100$$

B. FINANCIAL KPIs

1. Gross Profit (GP)

$$GP = \text{Revenue} - \text{Cost}$$

2. Gross Margin (GM)

$$GM = \frac{\text{Revenue} - \text{Cost}}{\text{Revenue}} \times 100$$

3. Profit Margin (PM)

$$PM = \frac{\text{Profit}}{\text{Revenue}} \times 100$$

4. Return on Investment (ROI)

$$ROI = \frac{\text{Profit}}{\text{Investment}} \times 100$$



COMMON ABBREVIATIONS

PV = Planned Value	GPI = Gravelizer
EV = Earned Value	CV = Schedule Variance
AC = Actual Cost	SV = Scheduling
BAC = Budget at Completion	TF = Total Float
EAC = Estimate at Completion	FF = Free Float
ETC = Estimate to Complete	PO = Purchase Order
VAC = Variance at Completion	

NOTES

- All costs should be in same currency.
- Time unit must be consistent (days/weeks/months).
- Use baseline (BAC & schedule) approved for comparison.
- Keep progress measurement method consistent.
- Update actuals regularly for reliable forecasts.



KEY PERFORMANCE AT A GLANCE



CPI > 1 → Under Budget

CPI = 1 → On Budget

CPI < 1 → Over Budget

SPI > 1 → Ahead of Schedule

SPI = 1 → On Schedule

SPI < 1 → Behind Schedule